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International Standard



6295

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Petroleum products — Mineral oils — Determination of interfacial tension of oil against water — Ring method

Produits pétroliers — Huiles minérales — Détermination de la tension interfaciale entre huile et eau — Méthode de l'anneau

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Foreword

ISO (the International Organization for Standardization) is a worldwide federation of national standards bodies (ISO member bodies). The work of developing International Standards is carried out through ISO technical committees. Every member body interested in a subject for which a technical committee has been authorized has the right to be represented on that committee. International organizations, governmental and non-governmental, in liaison with ISO, also take part in the work.

Draft International Standards adopted by the technical committees are circulated to the member bodies for approval before their acceptance as International Standards by the ISO Council.

International Standard ISO 6295 was developed by Technical Committee ISO/TC 28, *Petroleum products and lubricants*, and was circulated to the member bodies in February 1982.

It has been approved by the member bodies of the following countries :

Australia	Hungary	Portugal
Austria	India	Romania
Belgium	Iraq	South Africa, Rep.
Brazil	Israel	Spain
Bulgaria	Italy	Sri Lanka
Canada	Japan	Sweden
Czechoslovakia	Korea, Rep. of	Switzerland
Egypt, Arab Rep. of	Mexico	USA
France	Netherlands	USSR
Germany, F.R.	Poland	Venezuela

No member body expressed disapproval of the document.

Petroleum products — Mineral oils — Determination of interfacial tension of oil against water — Ring method

1 Scope and field of application

This International Standard specifies a method for the measurement, under non-equilibrium conditions, of the interfacial tension of mineral oils against water, which has been shown by practice to give a reliable indication of the presence of hydrophilic compounds.

2 Reference

ISO 3170, *Petroleum products — Liquid hydrocarbons — Manual sampling*.

3 Principle

Interfacial tension is determined by measuring the force necessary to detach a planar ring of platinum wire from the surface of the liquid of higher surface tension, that is, upward from the water-oil interface. To calculate the interfacial tension, the force so measured is corrected by an empirically determined factor which depends upon the force applied, the densities of both oil and water, and the dimensions of the ring. Measurements are made under rigidly standardized non-equilibrium conditions in which the measurement is completed within 1 min after formation of the interface.

4 Apparatus

4.1 Leconte du Nouy tensiometer, with platina ring of either 60 or 40 mm nominal circumference.

4.2 Ring, of fine platinum wire in a nearly true circle of 40 or 60 mm circumference welded to a suitable stirrup made of the same wire. It is necessary to know, to three significant figures, the circumference of each ring, and the ratio of the diameter of the ring to the diameter of the wire of which it is made.

4.3 Test portion container¹⁾, glass beaker or cylindrical vessel having a minimum diameter of 45 mm.

5 Preparation of apparatus

5.1 Clean all glassware by removing any residual oil with petroleum naphtha followed by several washes with methyl ethyl ketone and water and immersion in a hot cleaning solution of chromic acid. Rinse thoroughly with tap water and then with distilled water. Unless it is to be used immediately, drain the test portion container in an inverted position over a clean cloth.

5.2 Clean the platinum ring by rinsing it in petroleum naphtha followed by rinsing in methyl ethyl ketone. Then heat the ring in the oxidizing portion of a gas flame.

6 Calibration of apparatus

Make certain that all portions of the ring are in the same horizontal plane. Calibrate the tensiometer against known masses and adjust its zero point according to the procedure of its manufacturer.

7 Sampling

Carry out the sampling in accordance with one of the methods specified in ISO 3170.

8 Preparation of test portion

Filter the test portion through a 150 mm diameter, unwashed filter paper of medium porosity, using fresh paper for each 25 ml portion of oil.

9 Procedure

9.1 Determine the density of the test portion (clause 8) to the nearest 0,001 g/ml at 25 °C.

9.2 Introduce 50 to 75 ml of distilled water at a temperature of 25 ± 1 °C to a freshly cleaned test portion container (4.3)

1) Plasticizers used in the manufacture of tubing and containers can be detrimental to the quality of distilled water.

and place it on the adjustable platform of the tensiometer. Clean the platinum ring (4.2) and suspend it from the tensiometer (4.1). Raise the adjusting platform until the ring is immersed in the water to a depth not exceeding 6 mm in the centre of the test portion container as determined by visual observation.

9.3 Slowly lower the platform, maintain the torsion arm in the zero position by increasing the torque of the ring system. As the film of water adhering to the ring approaches the breaking point, proceed slowly with adjustment to ensure that the moving system will be in the zero position when rupture occurs. Using the scale reading at which this occurs, calculate the tension of the water test portion as specified in 10.1 using the value 0,997 g/ml for the difference of density of water and air ($\rho_0 - \rho_1$); a value of 71 to 72 mN/m should be obtained. If low values are found, they may be due to improper adjustment of the tensiometer or improperly cleaned apparatus. In this case, make the following readjustments: clean the test portion container with hot chromic acid cleaning solution, rinse, and repeat the measurement. If a low value is still obtained, further purify the distilled water (for example, by redistilling from an alkaline solution of potassium permanganate).

9.4 Return the tensiometer scale to zero and raise the adjustable platform until the ring is immersed in the distilled water to a depth of about 5 mm. Pour the filtered oil, previously brought to a temperature of $25 \pm 1^\circ\text{C}$, on the water to a depth of about 10 mm. Take care that the ring does not touch the oil-water interface. If a ring with a short stirrup is used, keep the oil level below the top of the stirrup to prevent bridging. If this is not possible, break the bridge with a suitable clean, sharp instrument as soon as possible after withdrawing the stirrup from the oil.

9.5 Allow the oil-water interface to age for 30 ± 1 s, then slowly lower the platform, maintaining the torsion arm in the zero position by increasing the torque of the ring system. As the water adhering to the ring approaches the breaking point, proceed slowly with adjustment to ensure that the moving parts will be in zero position when rupture occurs. Time these operations so that, as nearly as possible, 30 s are required to draw the ring through the interface. Proceed very slowly as the breaking point is approached, since the break is usually sluggish and too rapid movement may result in a high reading. Complete the entire operation, from the time of pouring the oil into the test portion container until the film ruptures, in about 1 min. Record the scale reading at which the ring breaks free from the interface.

10 Expression of results

10.1 Method of calculation

The interfacial tension, σ , expressed in millinewtons per metre (mN/m), is given by the formula

$$\sigma = M \times F$$

where

M is the calibrated scale reading of the tensiometer;

F is a factor calculated as follows :

$$F = 0,7250 + \sqrt{\frac{0,03678 \times M}{r_r^2(\rho_0 - \rho_1)} + P}$$

where

ρ_0 is the density of water, in grams per millilitre, at 25°C ;

ρ_1 is the density of the test portion, in grams per millilitre, at 25°C ;

P is a constant calculated as follows :

$$P = 0,04534 - \frac{1,679 \times r_w}{r_r}$$

where

r_w is the radius, in millimetres, of the wire of the ring,

r_r is the mean radius, in millimetres, of the ring.

10.2 Precision

The precision of the method, as obtained by statistical examination of inter-laboratory test results, is as follows :

10.2.1 Repeatability

The difference between successive test results, obtained by the same operator with the same apparatus under constant operating conditions on identical test material, would in the long run, in the normal and correct operation of the test method, exceed 2 % of the mean in only one case in 20.

10.2.2 Reproducibility

The difference between two single and independent results, obtained by different operators working in different laboratories on identical test material, would in the long run, in the normal and correct operation of the test method, exceed 5 % of the mean in only one case in 20.

11 Test report

The test report shall contain at least the following information :

- the type and identification of the product tested;
- a reference to this International Standard or to a national standard;
- the result of the test (see 10.1);
- any deviation, by agreement or otherwise, from the procedure specified;
- the date of the test.